RE07 Global Infrastructure Investment and Management

3. Private Investment in Infrastructure

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Lecture Outline

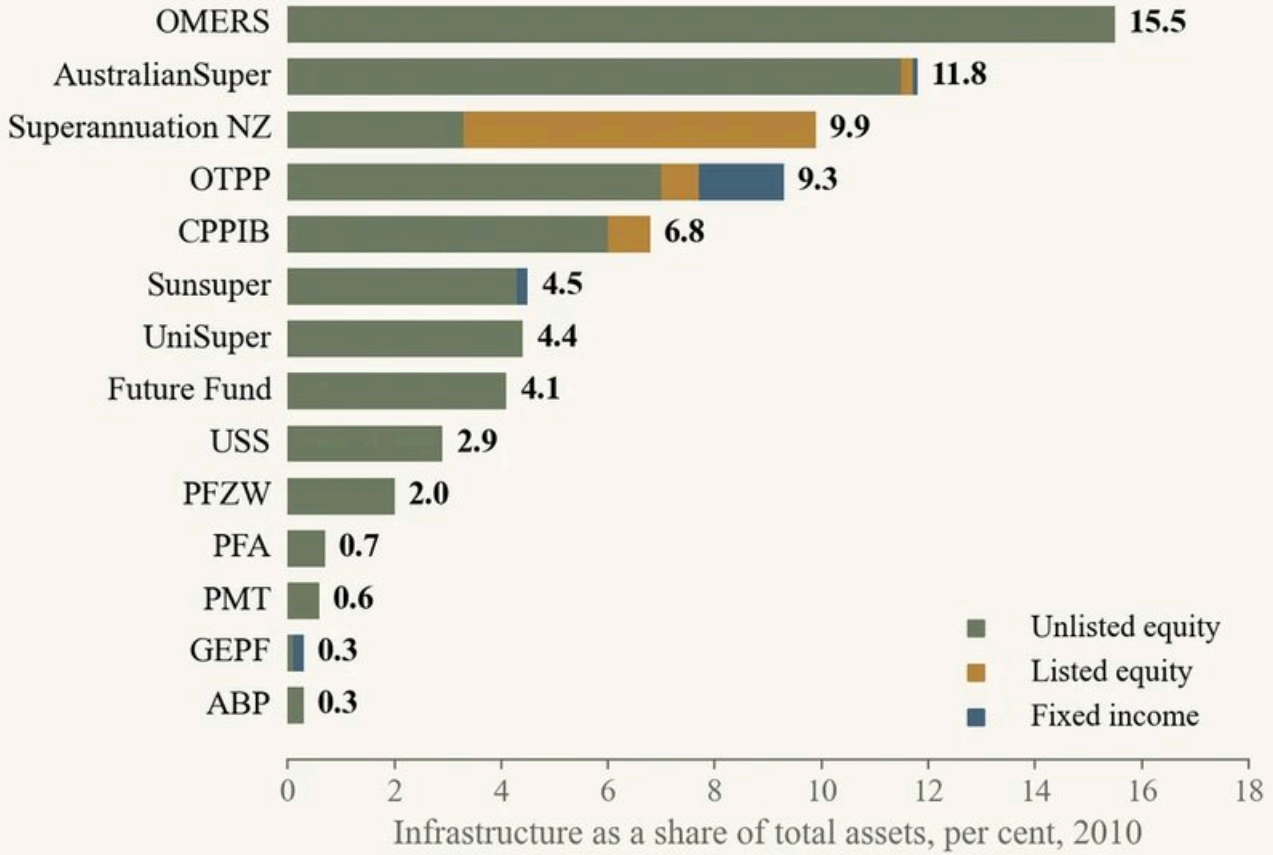
Investor types and their liabilities
The fund as the vehicle
Risk and return evidence
Project finance debt
Emerging markets and blended finance
Public or private ownership
Case Study: Newcore Capital and the Swansea Social Infrastructure Partnership (Hugo Llewelyn, Newcore Capital)

Investor Types and Their Liabilities

Investor	Liabilities and horizon	What it needs from the asset	Binding constraint
Pension funds	Decades, often linked to inflation	Income, inflation link, duration	Scale, governance capacity, valuation
Insurers and banks	Long for insurers; banks fund short	Predictable yield, low capital charge	Capital rules
Sovereign wealth funds	Few or none; very long horizon	Return and control at scale	Mandate and politics
Endowments and foundations	Perpetual, with a spending rule	Return and diversification	Liquidity and expertise
Development institutions	A mandate, not a required return	Additionality	Capital headroom

Della Croce, R. (2012), OECD Working Paper 29, page 7; Andonov, Kräussl and Rauh (2021), Review of Financial Studies 34(8), pages 3888 and 3909. Rows other than pension funds follow the earlier Week 3 deck.

Allocations Differ Widely across Pension Funds



Headline: 28 large funds held 2.9 per cent of their assets in infrastructure in 2010, mostly as unlisted equity

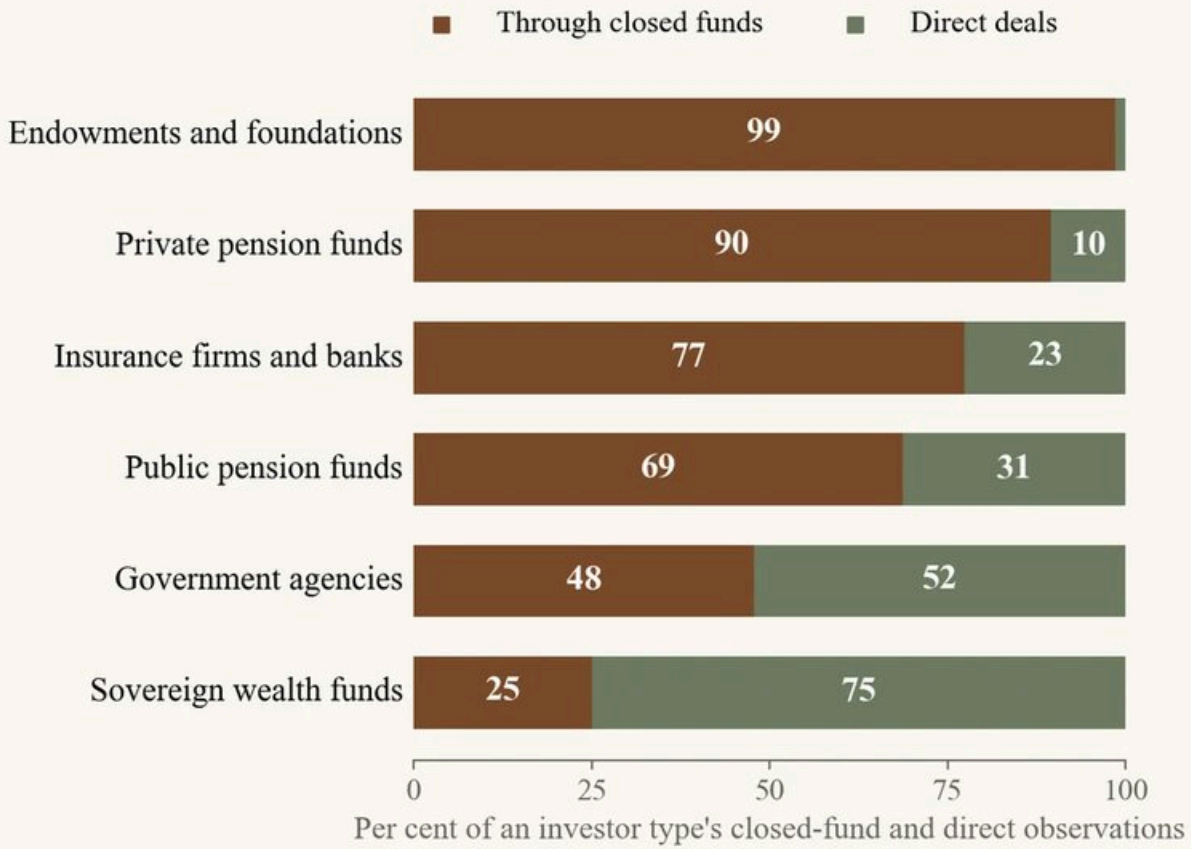
Spread: from 15.5 per cent at OMERS to 0.3 per cent at ABP

Route: of the unlisted equity, USD 30.9 billion was direct and USD 7.0 billion through funds

Caution: 2010 data for large funds; listed holdings sit inside equity allocations

Source: Della Croce, R. (2012), Trends in large pension fund investment in infrastructure, OECD Working Papers on Finance, Insurance and Private Pensions No. 29, Table 5 (page 23) and pages 4 and 8.

Routes Used by Institutional Investors



Closed funds: 538 funds are the main route for most investors

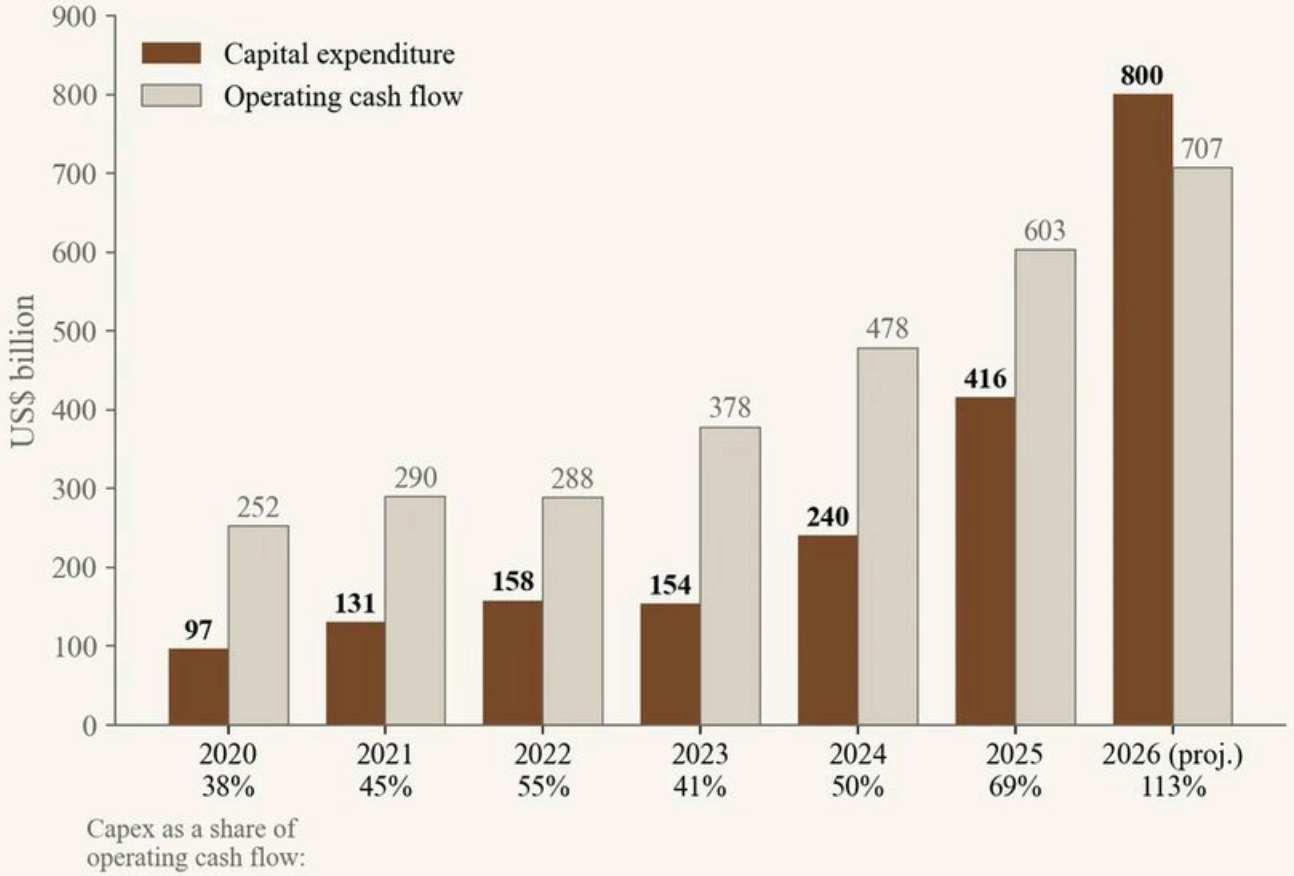
Direct deals: open to a few deep-pocketed investors, with the top 20 making 55 per cent of them

By investor type: sovereign funds and government agencies invest directly; endowments and private pension funds use funds

Reading: the shares compare investor-fund with investor-deal observations, not dollars

Andonov, Kräussl and Rauh (2021), Review of Financial Studies 34(8), Figure 1 and footnotes 2 and 6 (pages 3882 and 3888). Shares are the author's arithmetic on investor-fund (closed funds) and investor-deal (direct) observations.

Funding the Data Centre



Scale: 188 GW and about US$9 trillion over 2025 to 2032, 3.22 per cent of GDP a year

Cost: about US$41 million a megawatt: 11 for the building, 2 for power, 28 for chips and network

The gap: hyperscaler capex exceeds operating cash flow for the first time in 2026

The response: leases, joint ventures, project debt, private credit and securitisation

Van Nieuwerburgh, S. (2026), Financing the AI Buildout, NBER Working Paper 35865, Figure 1 and pages 2 to 8. Capex excludes data centres the hyperscalers lease; 2026 combines results, guidance and analysts' estimates.

The Capital Stack of a Data Centre

Layer	Who supplies it	Instrument	What it is exposed to
Tenant	Investment-grade hyperscalers	Internal cash flow, corporate debt, leases, guarantees	Demand for compute
Third-party equity	Developers, infrastructure funds, REITs, private equity	Equity in vehicles and joint ventures	Utilisation and residual value
Project debt	Banks and private credit funds	Mortgage loans, syndicated lines, private credit	Tenant credit and asset value
Structured finance	Bond and securitisation investors	Bonds from special-purpose vehicles backed by leases	Tenant concentration
IT equipment	Lessors and asset-backed investors	Leases and asset-backed securities	Obsolescence

Note: Hyperion (Meta and Blue Owl, 2025): about US$30 billion; 80 per cent of the equity sold for about US$2.5 billion; US$27 billion of debt rated A+, about 90 per cent of asset value, debt service cover 1.12, yield 6.58 per cent.

Van Nieuwerburgh, S. (2026), Financing the AI Buildout, NBER Working Paper 35865, pages 2 and 6–10. Hyperion figures are the paper's and are to be verified against the rating agency report before publication.

The Fund as the Vehicle

The structure: a limited partnership raised for 10 to 12 years, with capital called, invested and returned

When the money moves: calls early, distributions late, and a J-curve between them

Who is paid for what: a fixed fee and a share of the profit that behaves like an option

What the investor needs: an asset that moves with its liabilities, at a cost in liquidity

Andonov, Kräussl and Rauh (2021), Review of Financial Studies 34(8), page 3888; Metrick and Yasuda (2007), Wharton Rodney L. White Center Working Paper 17-07, page 2.

Takahashi and Alexander (2002): Cash Flows

Calls: a time-varying share of the commitment not yet called

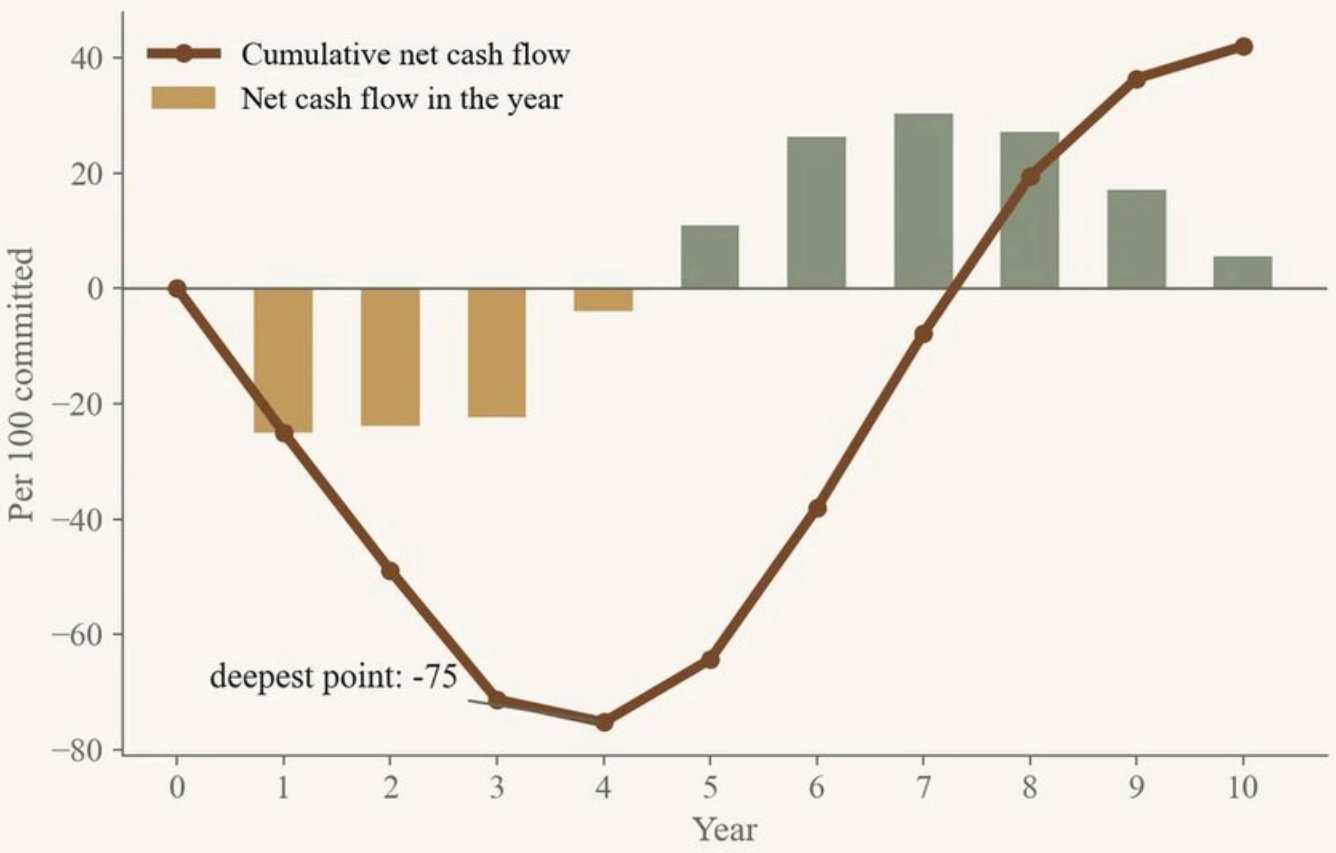
Distributions: a time-varying share of the grown asset value, never below the yield

Value: last year's value grown at G, plus calls, less distributions

Inputs: commitment, life, rate of contribution, bow factor, growth and yield

Takahashi, D. and Alexander, S. (2002), Illiquid alternative asset fund modeling, Journal of Portfolio Management 28(2), 90–100, as stated in Karatas, Klinkert and Hirsa (2021), arXiv 2108.02853, page 3.

The J-Curve of a Ten-Year Fund



Illustrative: net growth 9%, calls 25/33/50/50% of the uncalled commitment, bow factor 2.5, minimum yield 3%

Calls come first: the investor is a net payer for the first years

The J: the deepest cumulative outlay is about three quarters of the commitment, in year 4

Recovery: the cumulative flow turns positive in year 8 at 9 per cent net growth

Programme: annual commitments take years to build a stable portfolio

Author's calculation from the Takahashi and Alexander (2002) model as stated in Karatas, Klinkert and Hirsa (2021), arXiv 2108.02853, page 3; illustrative inputs (models_w3.py).

Metrick and Yasuda (2010): Fees as Claims

Fixed claim: the management fee is paid whatever the fund earns

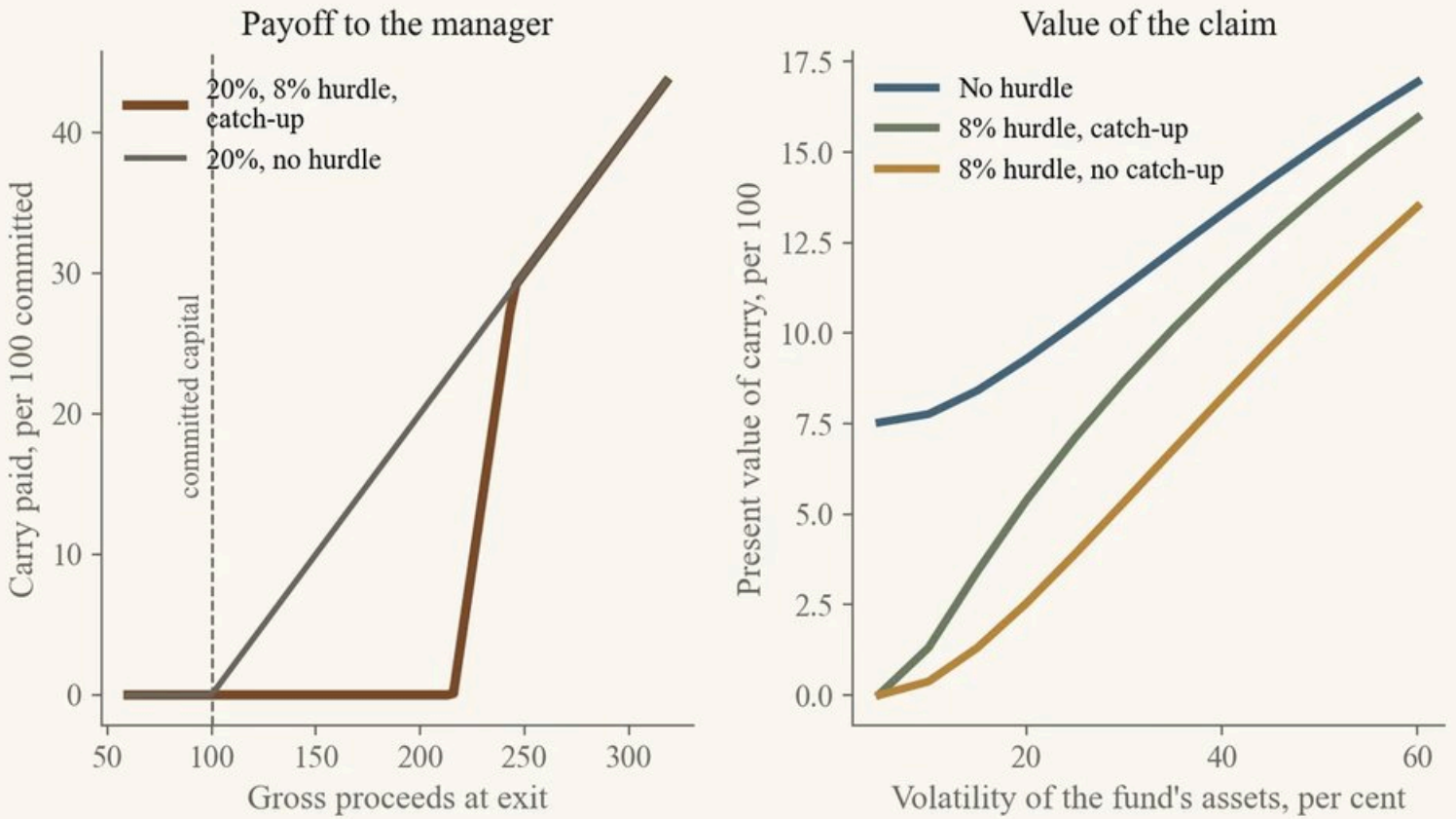
Variable claim: carried interest is a fraction of the profit and has the shape of a call option

Equilibrium: limited partners must earn their cost of capital after the manager is paid

Result: about 60 per cent of expected manager revenue is fixed

Metrick, A. and Yasuda, A. (2010), The economics of private equity funds, Review of Financial Studies 23(6), 2303–2341; Wharton Rodney L. White Center Working Paper 17-07 (2007), abstract and pages 2, 11–12 and 16.

The Value of Fixed and Variable Fees



Illustrative: 1.5% fee, 4% risk-free rate, 10 years

The shape: carry pays nothing below the hurdle, then rises with proceeds

The claim gains from risk: its value rises with volatility, as an option does

Default terms: fees 12.2 and carry 5.4 per 100 committed, a fixed share of 69 per cent

The price: the fund must beat its investable capital by about 1.9 per cent a year

Author's calculation after Metrick and Yasuda (2010): carry as a call option on a single exit, closed form (models_w3.py); 1.5% fee, 4% risk-free rate, 10 years. Illustrative.

Keel and Müller (1995): Assets against Liabilities

The problem: maximise expected surplus return less half its variance, where surplus is assets less liabilities

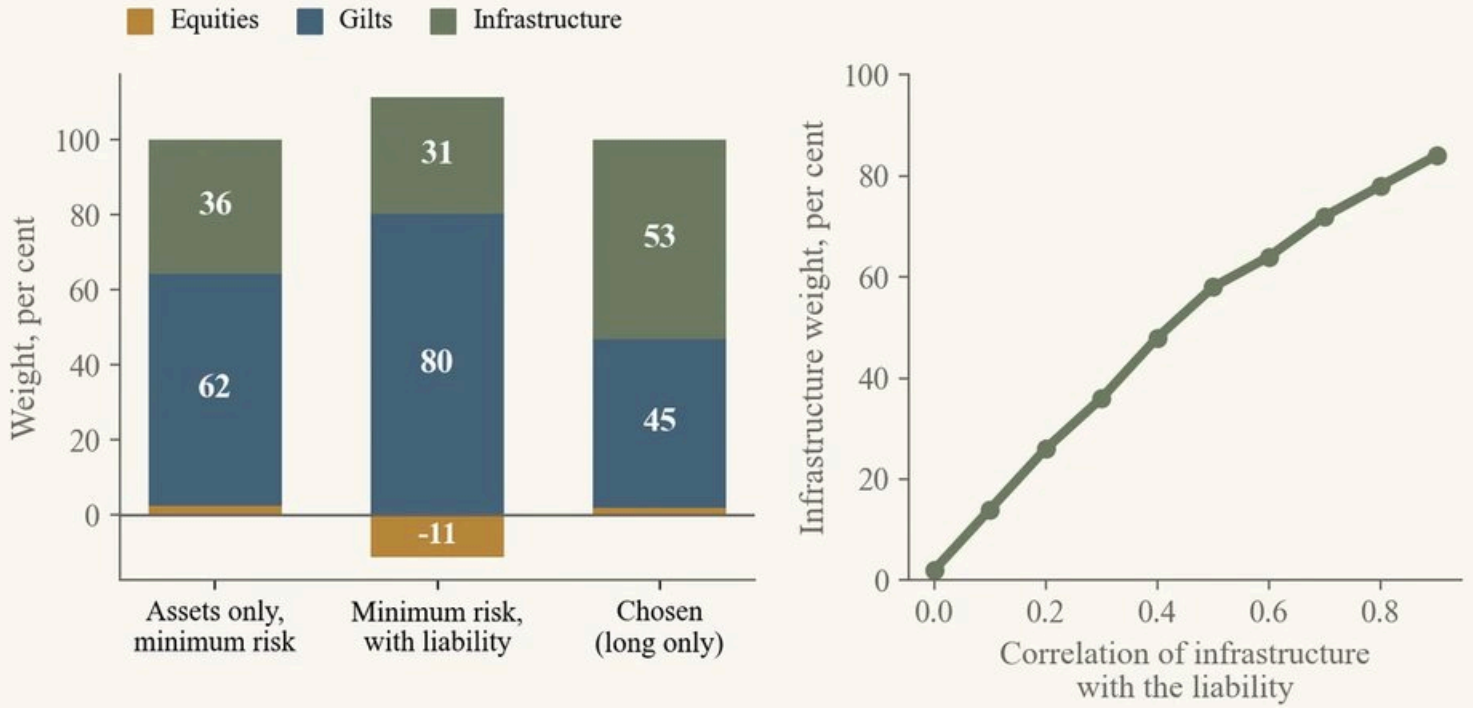
The result: the liability shifts the minimum-risk portfolio, whatever the investor's risk tolerance

The test: an asset earns its place by its correlation with the liability, not only by its own risk

The caveat: one period, no illiquidity, and the correlation is an assumption

Keel, A. and Müller, H. H. (1995), Efficient portfolios in the asset liability context, ASTIN Bulletin 25(1), 33–48, pages 33–39; after Sharpe, W. F. and Tint, L. G. (1990), Journal of Portfolio Management 16(2), 5–10.

Hedging the Liability with Real Assets



Illustrative: infrastructure 6.5% return, 11% volatility; risk tolerance 0.2; fully funded

Hedge portfolio: counting the liability moves the minimum-risk mix from 36 to 31 per cent infrastructure, and shorts equities

Chosen portfolio: at a risk tolerance of 0.2, 53 per cent infrastructure and 45 per cent gilts

The link matters: the weight runs from 2 to 84 per cent as the correlation with the liability rises from 0 to 0.9

Illiquidity: the cost the asset can bear is 2.6 per cent a year at the default settings

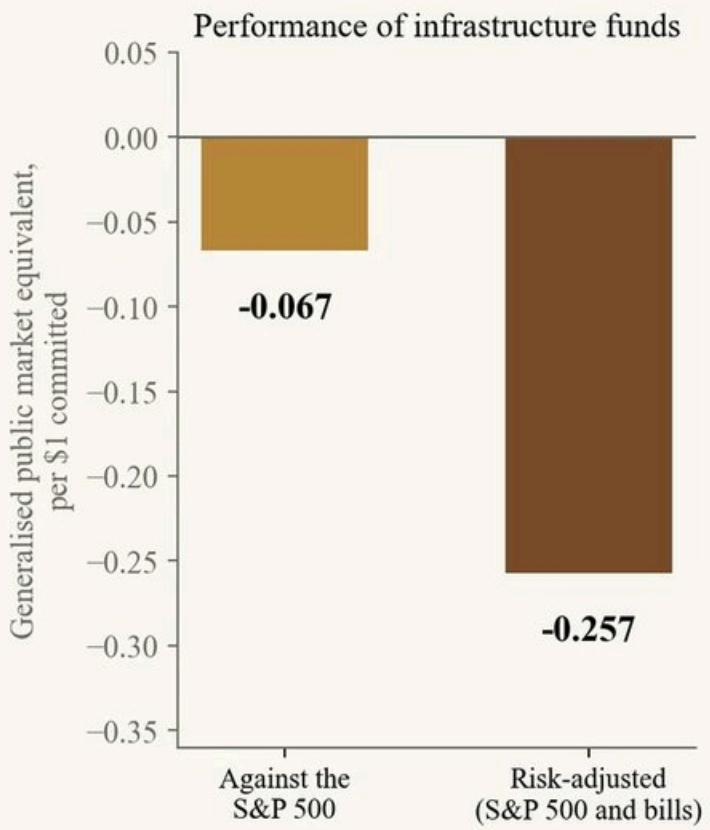
Author's calculation after Keel and Müller (1995), ASTIN Bulletin 25(1), pages 35–39 (models_w3.py); illustrative returns, volatilities and correlations; risk tolerance 0.2, funding ratio 1.

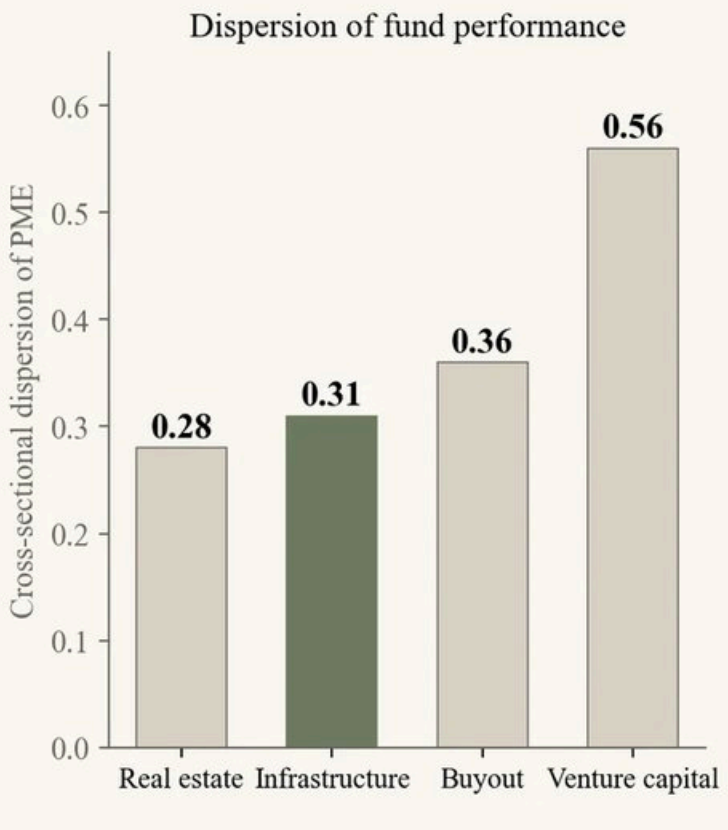
Three Models of the Fund Compared

Model	Explains	Prediction	Evidence and limit
Takahashi and Alexander (2002)	When the money moves	Calls early, returns late: a J-curve and a mid-life exposure peak	Standard planning tool; deterministic, silent on fees and return
Metrick and Yasuda (2010)	Who is paid for what	Carry is an option, worth more with risk; most revenue is fixed	238 funds, about 60 per cent fixed; one investor's data
Keel and Müller (1995)	What the investor needs	An asset earns its place by its link to the liability	Theory; one period; funds look like other private funds

Karatas, Klinkert and Hirsa (2021), page 3; Metrick and Yasuda (2007), abstract and page 16; Keel and Müller (1995), pages 38–39; Andonov, Kräussl and Rauh (2021), pages 3882–3883.

Risk and Return of Unlisted Infrastructure Funds





Average: a public market equivalent of 0.93, below buyout, venture capital and real estate

Risk-adjusted: a loss of 0.257 per dollar committed, comparable to venture capital

Dispersion: 0.31, similar to buyout and real estate, below venture capital

Leverage: the median infrastructure deal is financed 66 per cent by debt

Andonov, Kräussl and Rauh (2021), Review of Financial Studies 34(8), Table 6 (page 3908) and pages 3882–3883 and 3909. GPME from Korteweg and Nagel's method; funds raised 2002–2018.

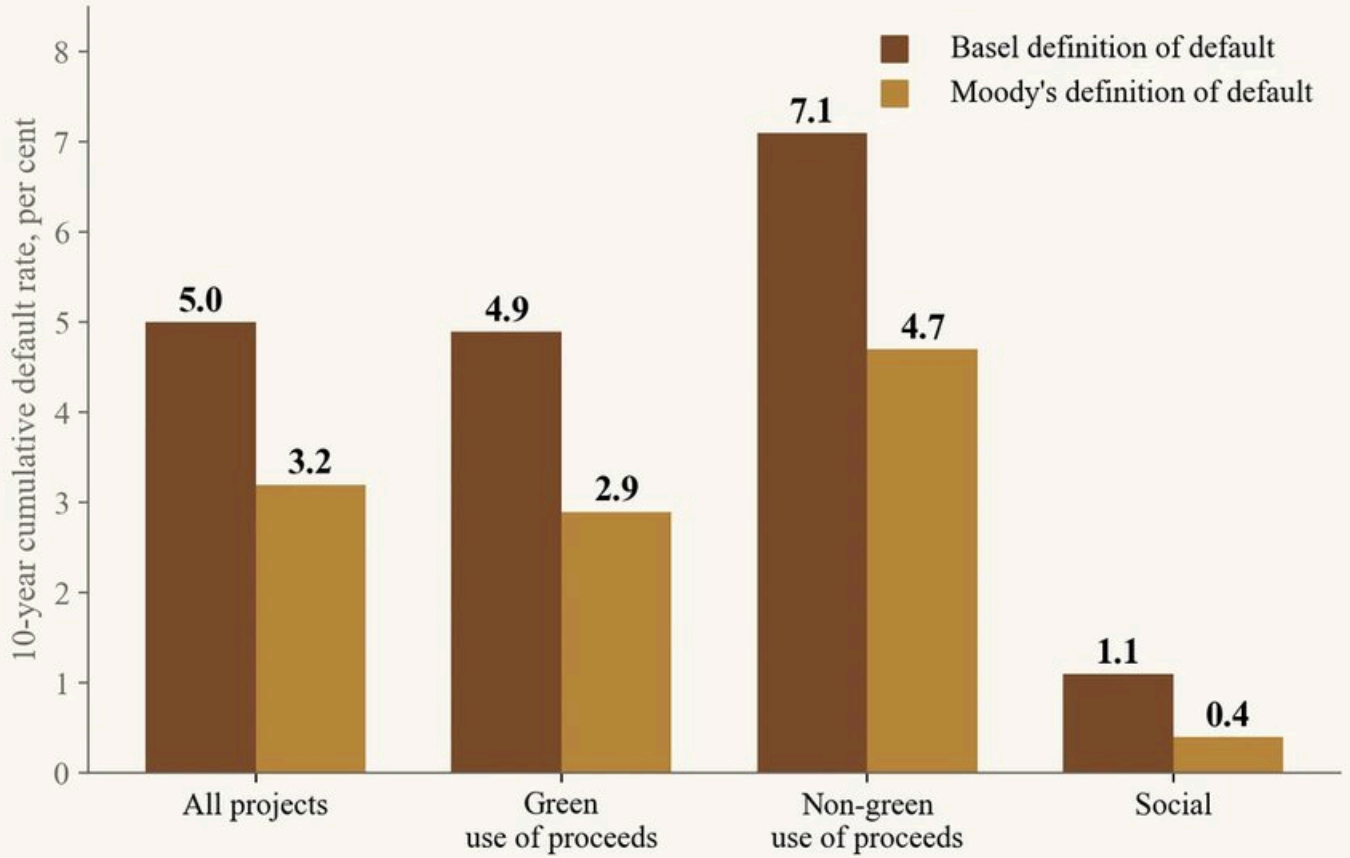
Exits, Cash Flows and Investor Type

Association with performance	Public market equivalent	Net IRR, percentage points	Multiple of invested capital
10 percentage points more of deals exited in years 0 to 5	+0.085	+2.575	+0.079
A public investor, compared with a private investor	−0.026	−1.810	−0.038

Closed funds are paid to exit quickly, and public investors in them earn less, after controlling for deal stage, region and industry.

Andonov, Kräussl and Rauh (2021), Review of Financial Studies 34(8), Table 4 (page 3899), Table 8 (page 3915) and pages 3883–3884.

Default and Recovery in Project Finance Loans



Default: 5.0 per cent of project loans default within ten years on the Basel definition, 3.2 per cent on Moody's

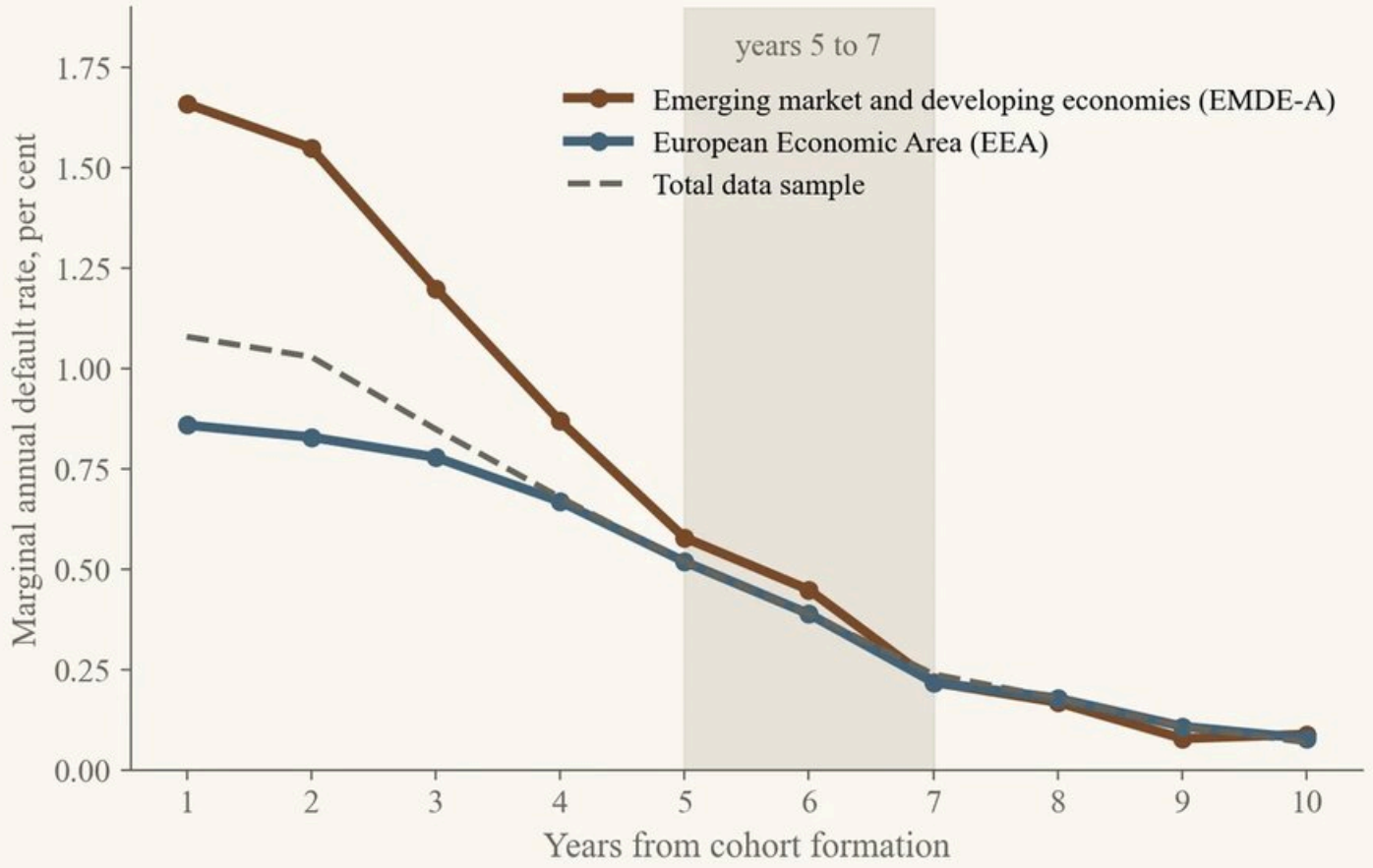
Recovery: the average recovery is 81.5 per cent, and the median is 100 per cent

The pattern: contract terms and offtake explain more than sector labels

Caution: bank-loan data from a consortium; the March 2020 study was not read

Moody's Investors Service (2020), Default and recovery rates for project finance bank loans, 1983–2018: Sustainable project finance bank loans, Exhibit 2 (page 8) and pages 1–7. Total data sample of 7,354 projects.

Risk Falls with an Operating Record



Seasoning: default risk is highest in years 1 to 4 and falls steadily

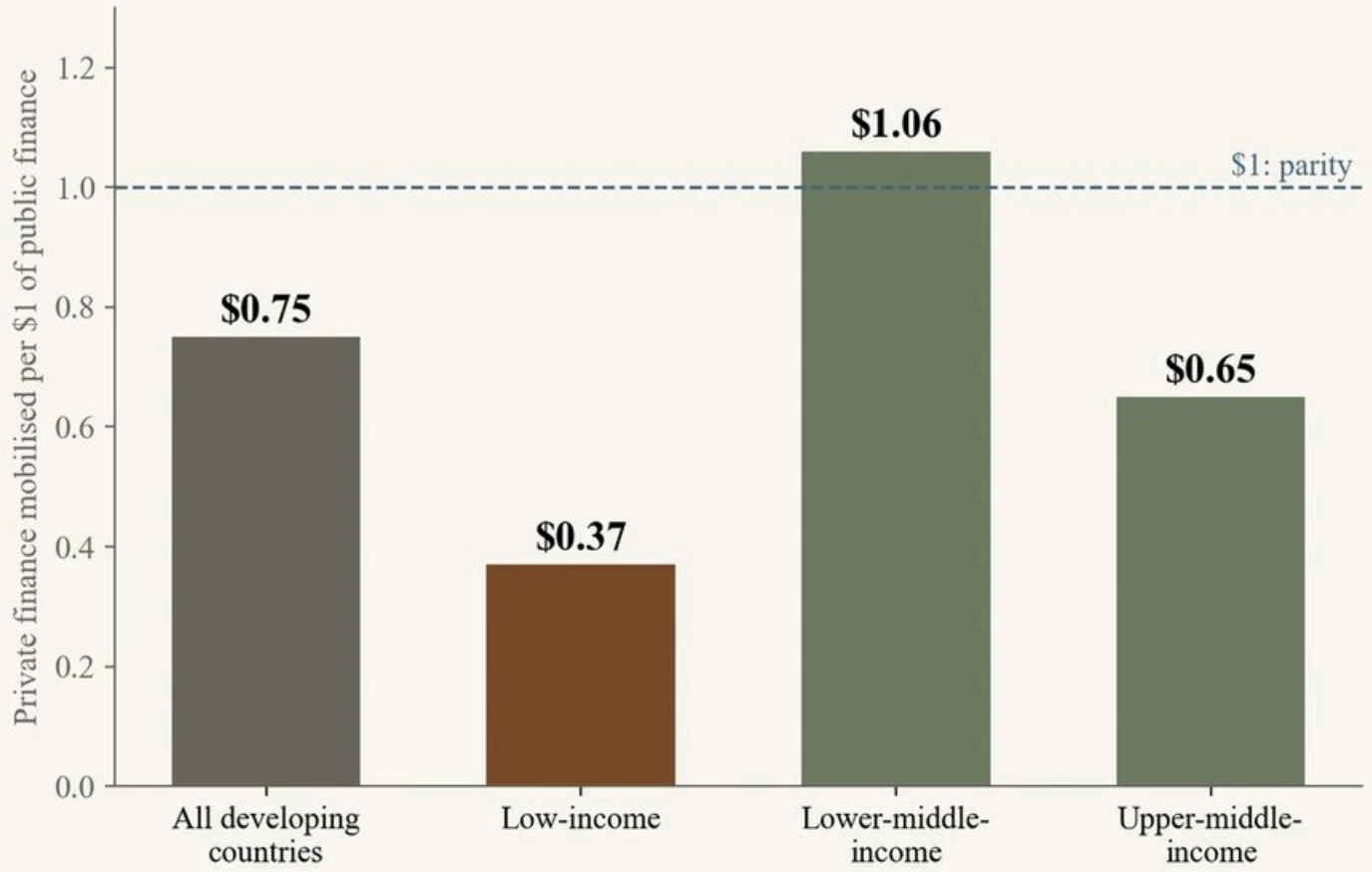
Convergence: by year 7 regions and uses of proceeds look alike, near single-A corporates

Causes: market risk in Europe; country risk in emerging markets

Entry point: buying a seasoned loan buys a lower-risk stage of the same asset

Moody's Investors Service (2020), Default and recovery rates for project finance bank loans, 1983–2018, Exhibit 23 (page 25) and pages 1 and 7. Basel definition of default.

Blended Finance: Leverage Ratios



Overall: each US$1 of public commitment mobilised US$0.75 of private finance, 2013 to 2015

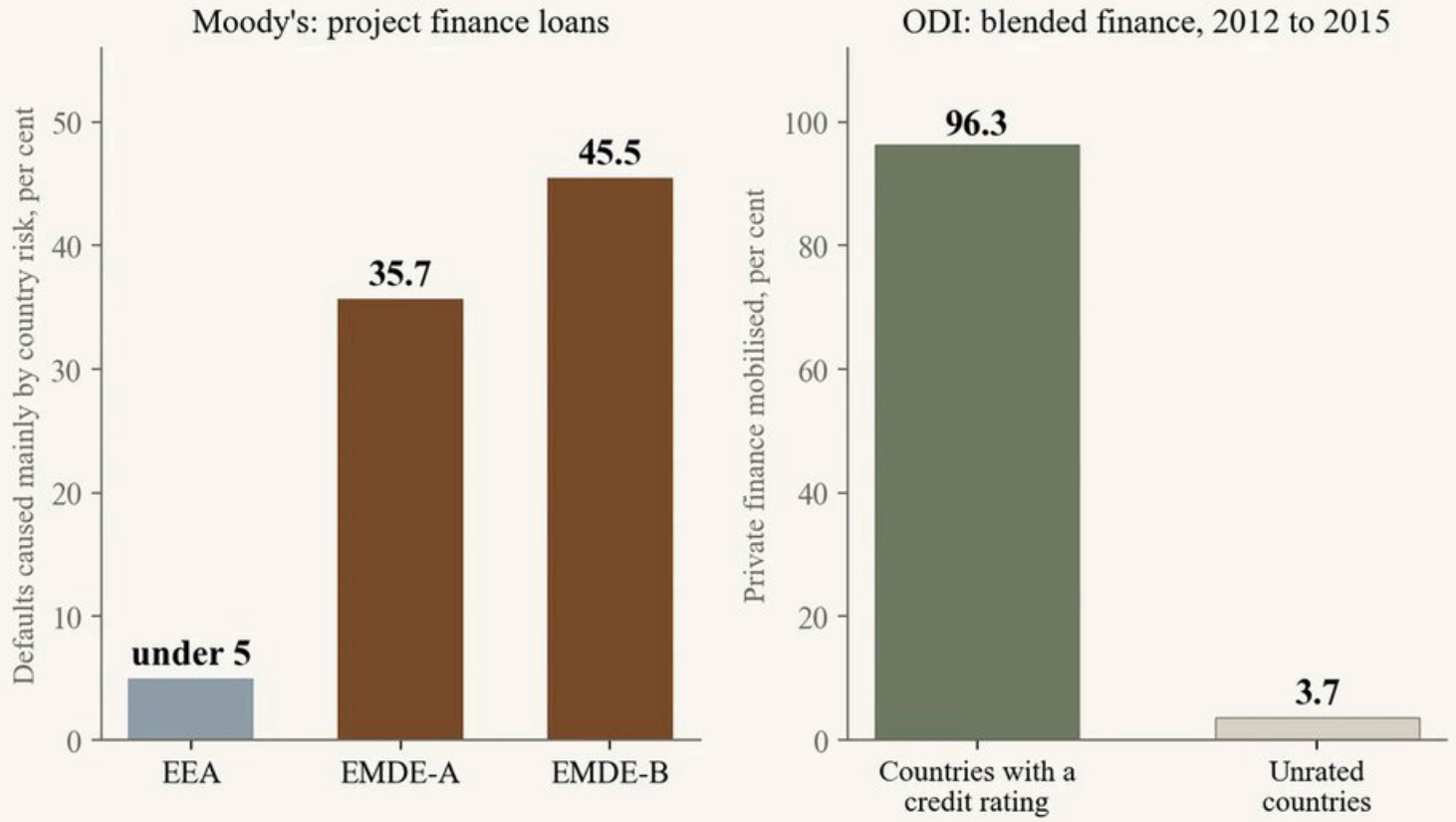
Low-income countries: US$0.37, rising to US$1.06 in lower-middle-income countries

Infrastructure: the lowest ratios in low- and middle-income countries

Reading: ‘billions to billions’ is more plausible than ‘billions to trillions’

Attridge, S. and Engen, L. (2019), Blended finance in the poorest countries: the need for a better approach, Overseas Development Institute, key findings (page 11) and pages 37–38. Nine selected MDBs and DFIs, 2013–2015.

Private Finance Follows Credit Ratings



Ratings: 96.3 per cent of mobilised private finance went to countries with a credit rating

Default: country risk caused 36 to 46 per cent of emerging-market project defaults, and under 5 per cent in Europe

Reading: instruments cannot supply the invdfestment climate

Implication: enforceable contracts and priced risk come first

Attridge, S. and Engen, L. (2019), ODI, pages 11 and 53; Moody's Investors Service (2020), page 7. Left panel: share of defaults caused mainly by country risk; EEA drawn at the bound of 5 per cent.

Megginson and Netter: The Survey Evidence

The claim: privately owned firms outperform state-owned firms, and privatisation improves operating and financial performance

The scale: 75 of the Global 1000 were privatised firms in 1998; governments raised over US$500 billion in share offerings

A welfare study: Galal and co-authors: gains in 11 of 12 divestitures, about 26 per cent of sales

The condition: public ownership wins only where quality cannot be contracted and competition is weak

Megginson, W. L. and Netter, J. M. (2001), From state to market: a survey of empirical studies on privatization, Journal of Economic Literature 39(2), 321–389; conference draft of December 1998, pages 2, 8–9 and 12–13 and the appendix summary table.

Two Readings of Private Provision

	Megginson and Netter	Bel, Fageda and Warner
Scope	State-owned enterprises in many industries, mostly large firms	Solid waste collection and water distribution
Method	Survey of before-and-after and comparative studies	Meta-regression of 27 econometric studies
Finding	Private ownership improves performance	No systematic cost saving from private production
What reconciles them	Competition and regulation accompany the gains	Savings depend on competition, contract design and asset specificity

Megginson and Netter (2001), draft of December 1998, abstract and pages 12–13; Bel, Fageda and Warner (2009), IREA Working Paper 2009/23, abstract and pages 2–3 and 23. Journal version: Journal of Policy Analysis and Management (2010), not retrieved.

Case Study: Newcore Capital and the Swansea Social Infrastructure Partnership (Hugo Llewelyn, Newcore Capital)

The fund and its partners: who set it up, and who invested

The investor's problem: what the investor needed

Terms, returns and risks: the structure of the fund and its fees

Measuring outcomes: what the partnership set out to deliver

Discussion: questions for the guest

Summary

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